

Assignment 2 – Q1 Solution (SVM)

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We are given three training samples:

$$\begin{aligned}\mathbf{x}_1 &= (1, 1)^T, \text{ label } y_1 = +1 \\ \mathbf{x}_2 &= (2, 2)^T, \text{ label } y_2 = +1 \\ \mathbf{x}_3 &= (0, 0)^T, \text{ label } y_3 = -1\end{aligned}$$

The soft-margin SVM uses a quadratic polynomial kernel:

$$k(\mathbf{x}_i, \mathbf{x}_j) = (\mathbf{x}_i^T \mathbf{x}_j + 1)^2$$

Regularization parameter: $C = 1.0$

The obtained Lagrange multipliers are:

$$\alpha_1 = 0.25, \quad \alpha_2 = 0, \quad \alpha_3 = 0.25$$

Identify the Support Vectors

Support vectors satisfy $\alpha_i > 0$.

Thus:

$\mathbf{x}_1$ and $\mathbf{x}_3$ are support vectors.

Compute the Bias Term b

For any support vector with $0 < \alpha_i < C$:

$$\sum_{j=1}^3 \alpha_j y_j k(\mathbf{x}_j, \mathbf{x}_i) + b = y_i$$

Using $\mathbf{x}_1$:

$$k(\mathbf{x}_1, \mathbf{x}_1) = 9, \quad k(\mathbf{x}_2, \mathbf{x}_1) = 25, \quad k(\mathbf{x}_3, \mathbf{x}_1) = 1$$

$$\sum \alpha_j y_j k(\mathbf{x}_j, \mathbf{x}_1) = 0.25(1)(9) + 0 + 0.25(-1)(1) = 2.25 - 0.25 = 2.0$$

$$2.0 + b = 1 \quad \Rightarrow \quad b = -1$$

Expand the Decision Function as a Polynomial

Let $\mathbf{x} = (x_a, x_b)^T$.

$$k(\mathbf{x}_1, \mathbf{x}) = (x_a + x_b + 1)^2$$

Contribution from $\alpha_1 y_1$:

$$0.25(x_a^2 + x_b^2 + 2x_a x_b + 2x_a + 2x_b + 1)$$

Contribution from $\alpha_2 = 0$ is zero.

Contribution from $\alpha_3 y_3 = -0.25$:

$$-0.25$$

Add bias $b = -1$.

Final polynomial:

$$0.25x_a^2 + 0.25x_b^2 + 0.5x_a x_b + 0.5x_a + 0.5x_b - 1$$

Classify the New Sample $(1, 0)^T$

Substitute $x_a = 1, x_b = 0$:

$$0.25(1) + 0 + 0 + 0.5(1) + 0 - 1 = -0.25$$

Decision:

$$\text{sign}(-0.25) = -1$$

Thus the predicted label is -1 .